Chapter 4

扩散

Diffusion of dopants

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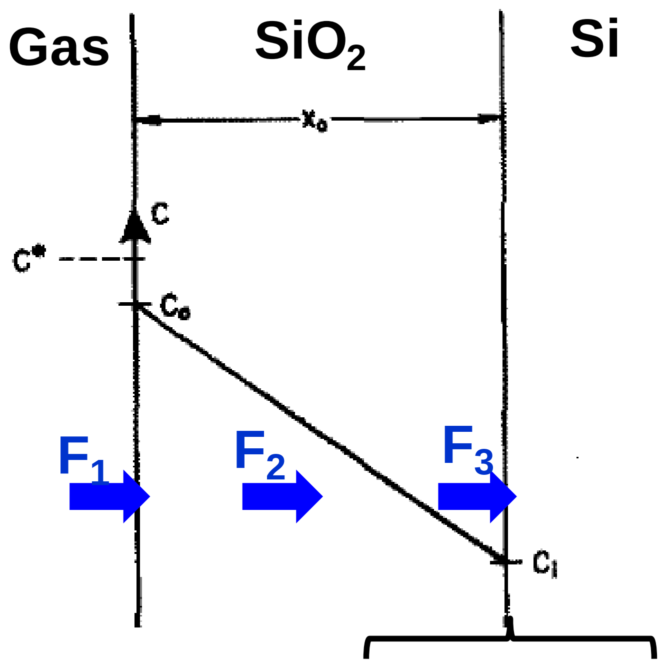
2025年11月20日

Outline

- 4. 1 Why diffusion is important?
- 4. 2 Basics of Diffusion
- 4. 3 Fick's law and solution for diffusion
- 4. 4 High concentration diffusion effect
- 4. 5 Other aspect of diffusion
- 4.6 Summary and homework

4.1Why is diffusion important

F: flux – the number of oxygen molecules that crosses a plane per unit area per second



O_2+Si↔SiO_2

Si+2HO_2→SiO_2 +H_2

At surface

$$F_1 = h(C^* - C_o)$$ h_g是质量输运系数

h: Mass transfer coefficient [cm/sec].

In SiO_2 **Deff: Diffusivity [cm^2/sec]**

$$F_2 = D_{eff}\frac{(C_0 - C_i)}{x}$$ 菲克第一定律

At SiO_2/Si interface

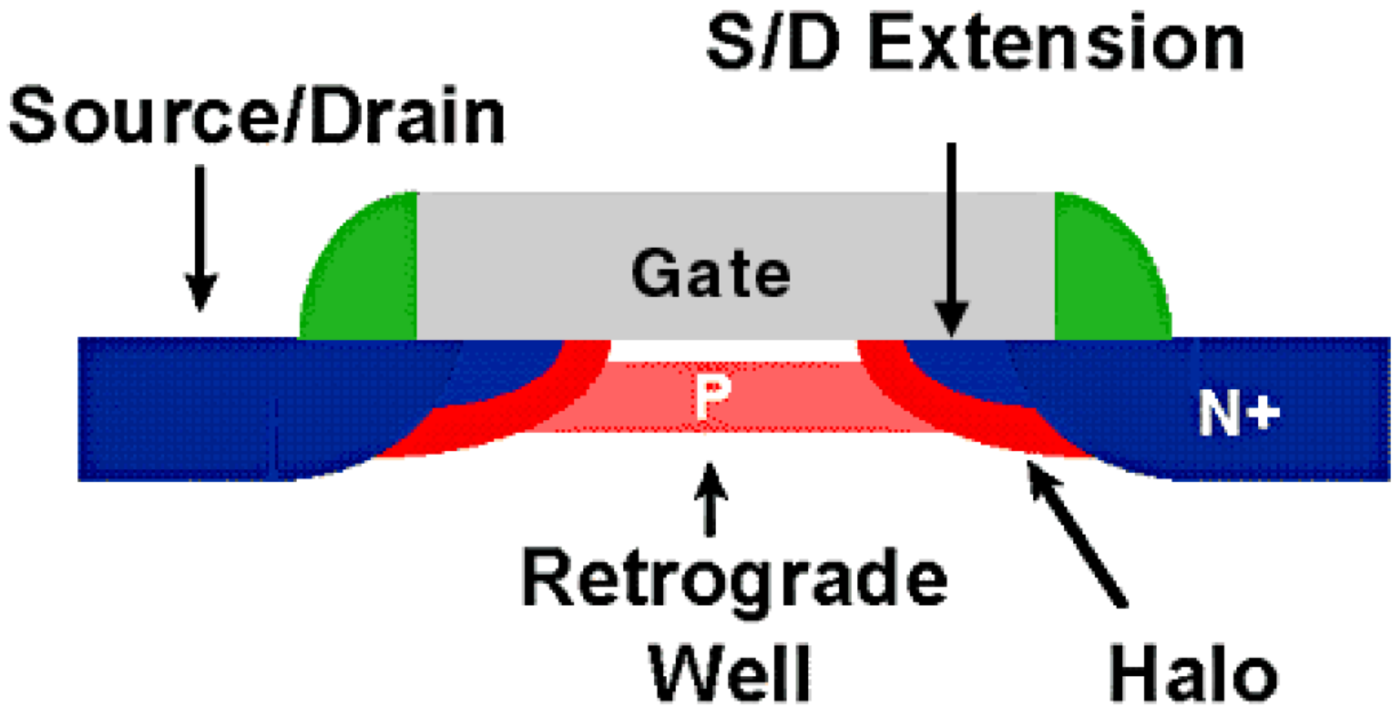
$$F_3 = kC_i$$

k: Oxidation reaction rate constant

Oxidant diffusion→ interface reaction.

4.1Why is diffusion important?

Strategies to overcome short channel effect

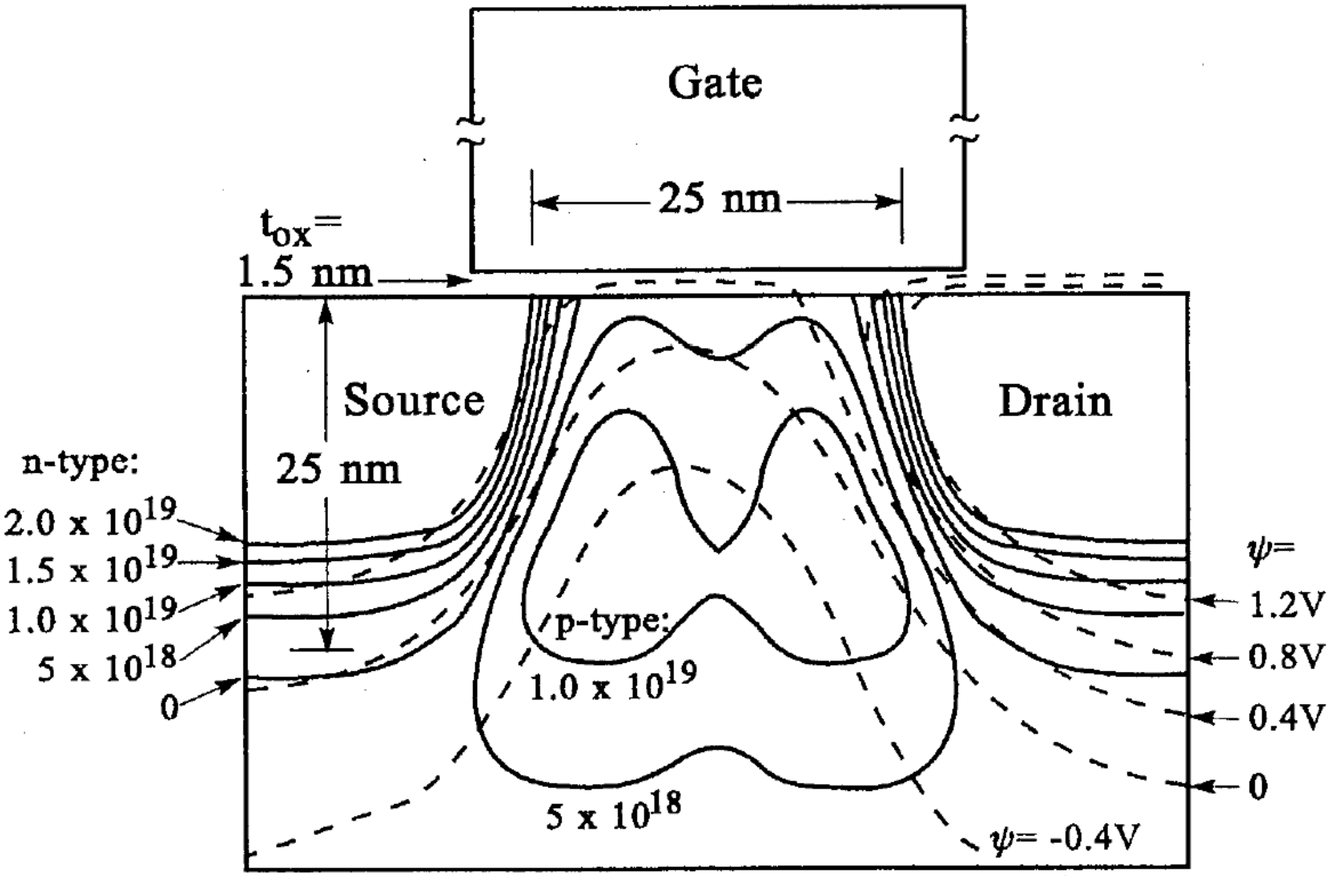


- Anti-punchthrough implantation（PTI）
- Halo implantation

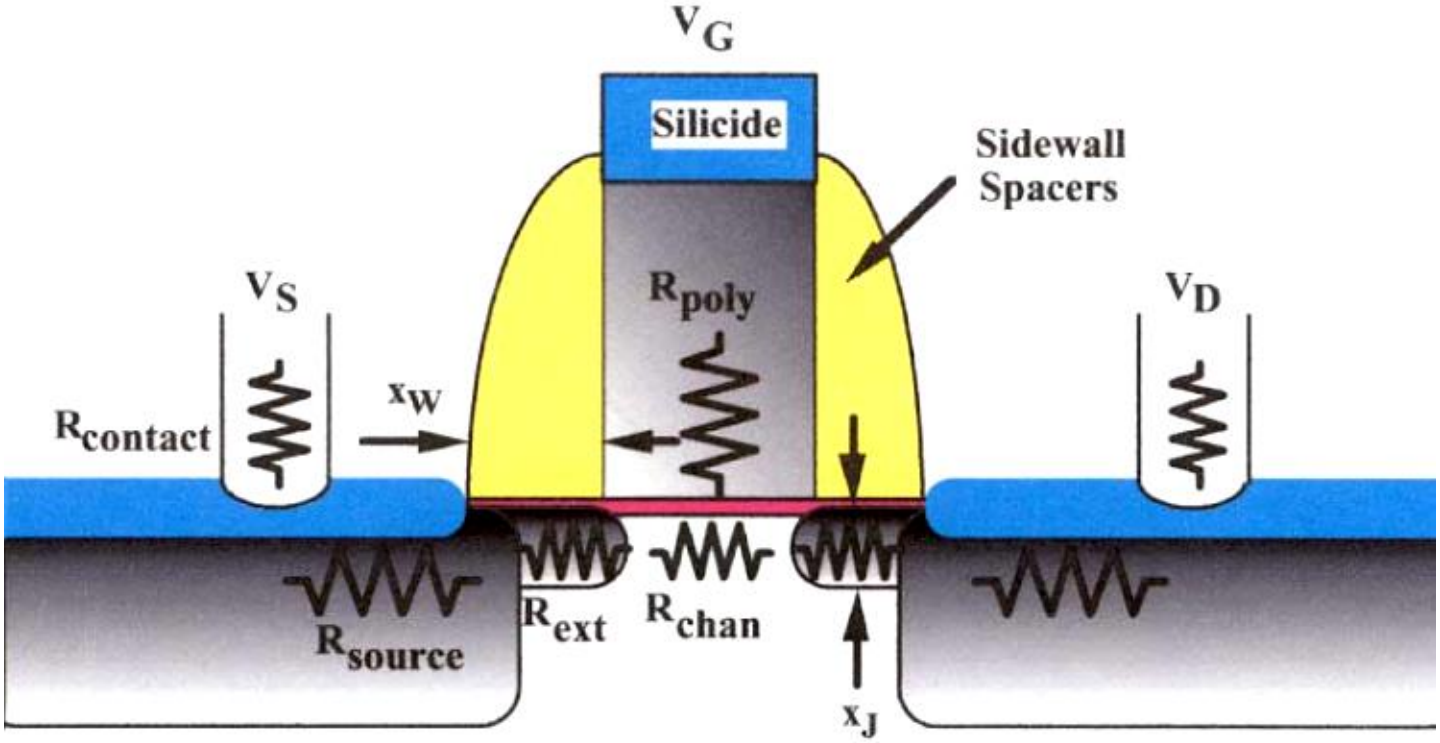
4.1Why is diffusion important?

Halo or Pocket Profiles: Improve Manufacturability

25 nm Gate Length MOSFET Design



4.1Why is diffusion important?



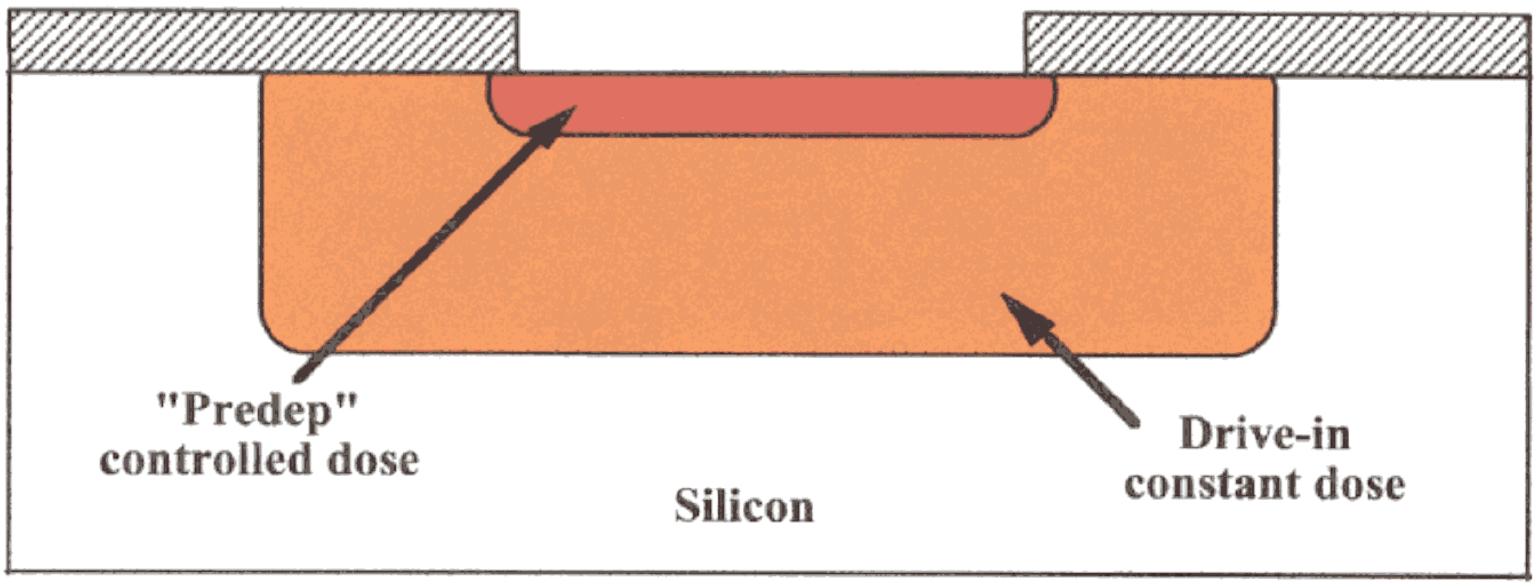
Technology requirements demand knowledge of dopant positions with almost atomic-scale accuracy, in 2D and 3D profiles.

It is based on the knowledge of diffusion

Outline

4. 2.1 How does diffusion occur?

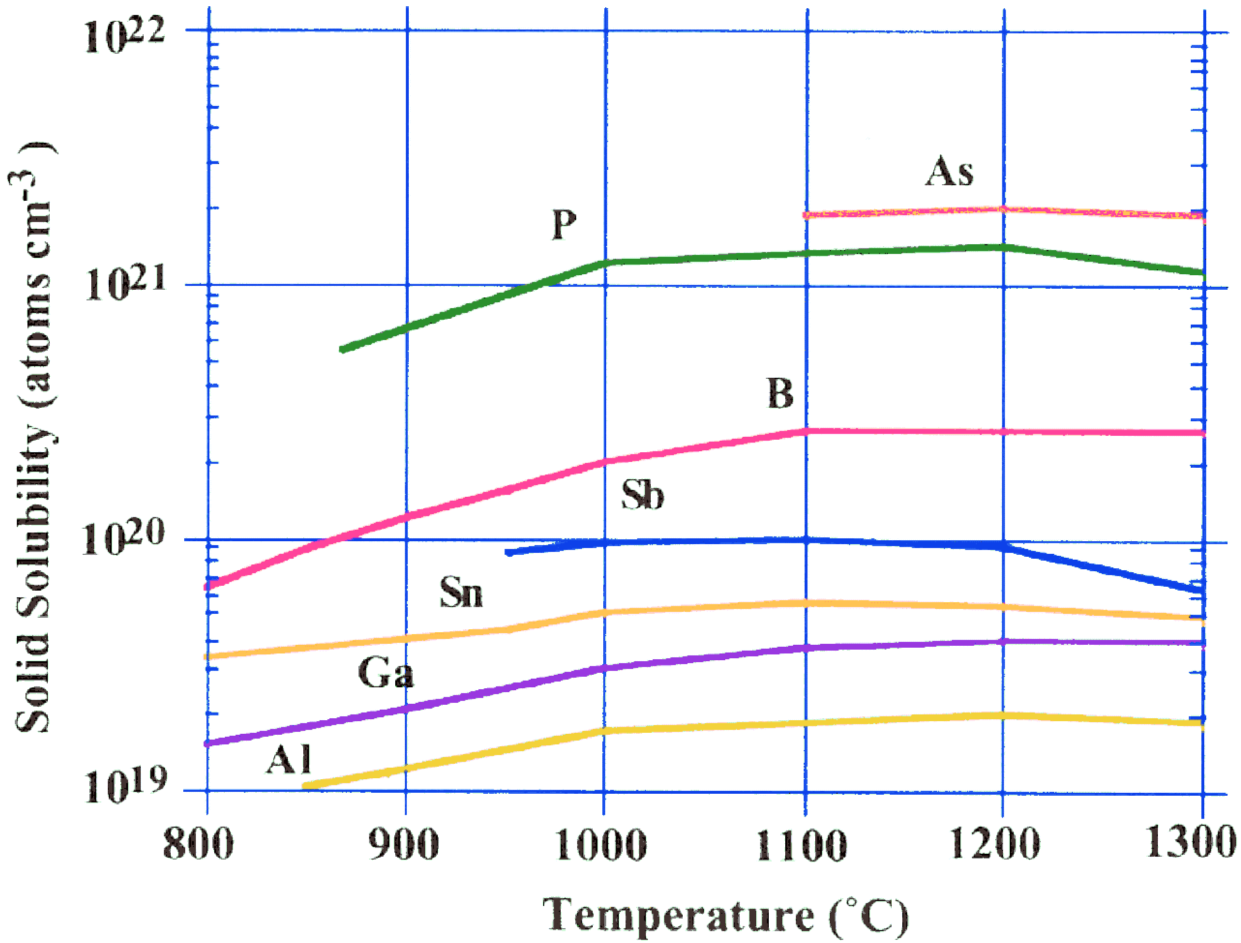
Diffusion is the redistribution of atoms from regions of high concentration of mobile species to regions of low concentrations



Historical development: original method of introducing dopants into the silicon crystal was by doing a gas-phase diffusion (high temperature) masked by oxide. This method has serious limitations on the control of the dose of dopant introduced, as well as the profile shape.

Modern method: use ion implantation to introduce dopants, and annealing/diffusion to repair damage and, if necessary, diffuse the profile.

4. 2.1 Solid solubility of dopants

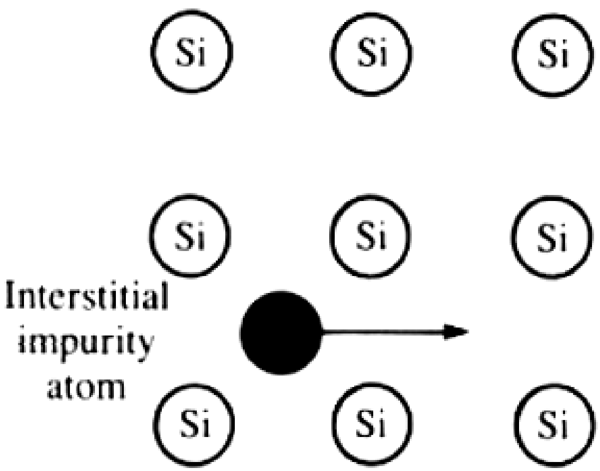


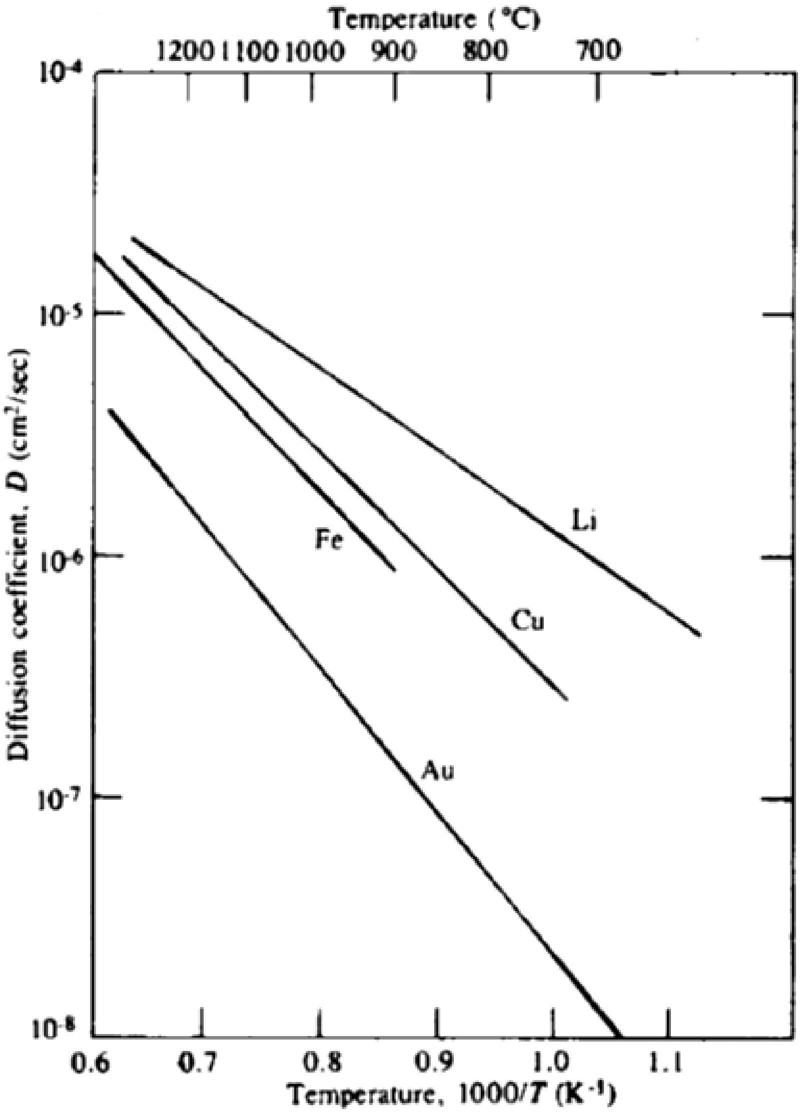
Dopants are soluble in bulk silicon up to a maximum value before they precipitate into another phase ("solid solubility").

4.2.2 Diffusion mechanisms in Si

(1) Interstitial Diffusion

Example: Cu, Fe, Li, H

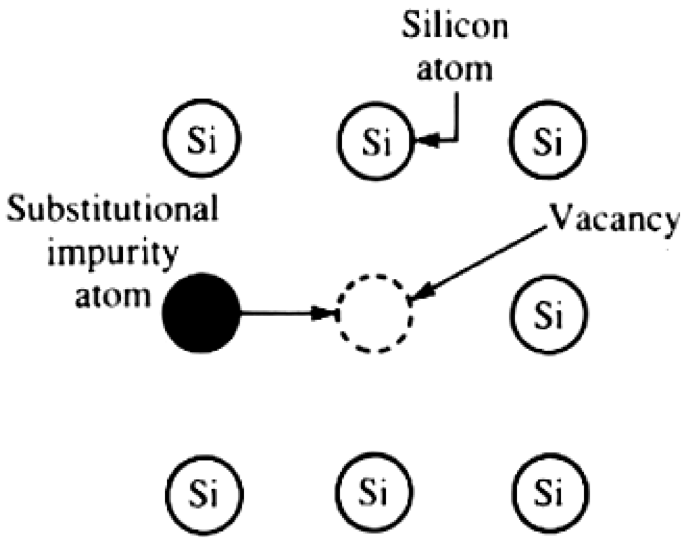


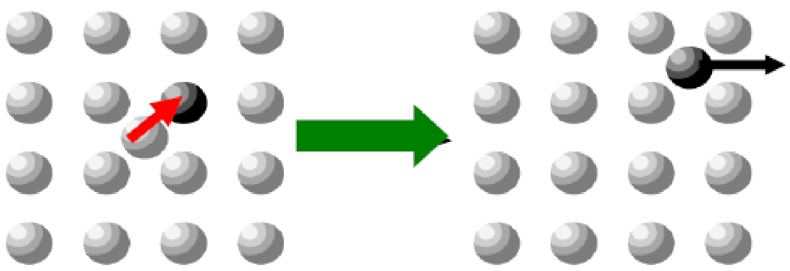


4.4.2 Diffusion mechanisms in Si

(b) Substitutional Diffusion (c) Interstitialcy Diffusion

Example: Dopants in Si (e.g. B, P,As,Sb)





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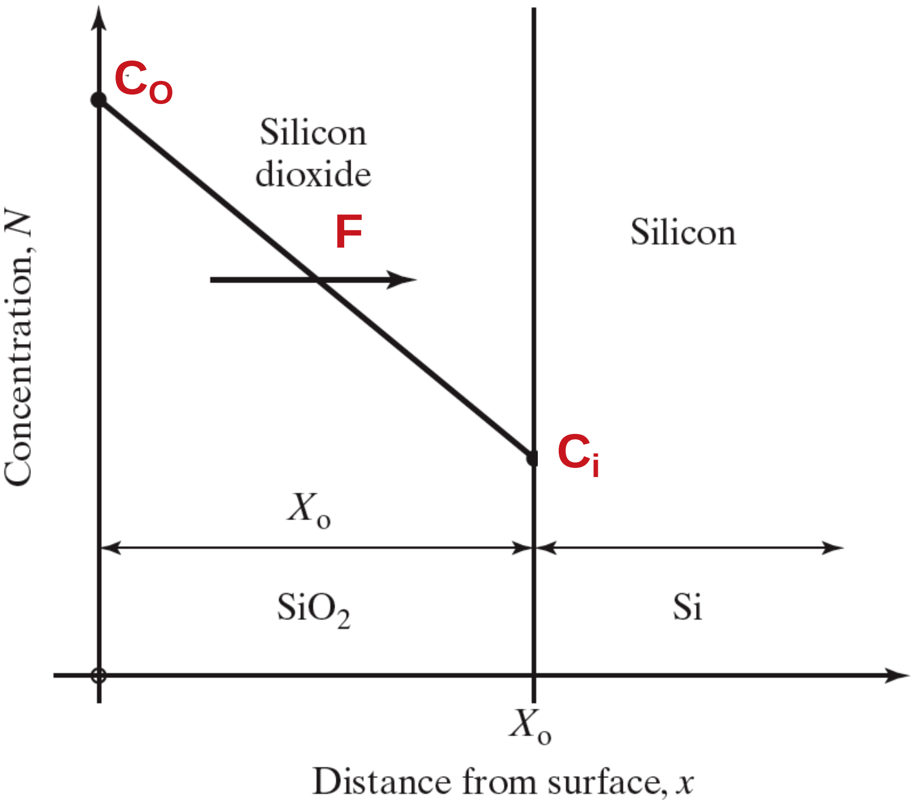
Fick's first law:

How the flux of dopant depends upon the doping gradient:



$$F = -\frac{DdC}{dx}$$

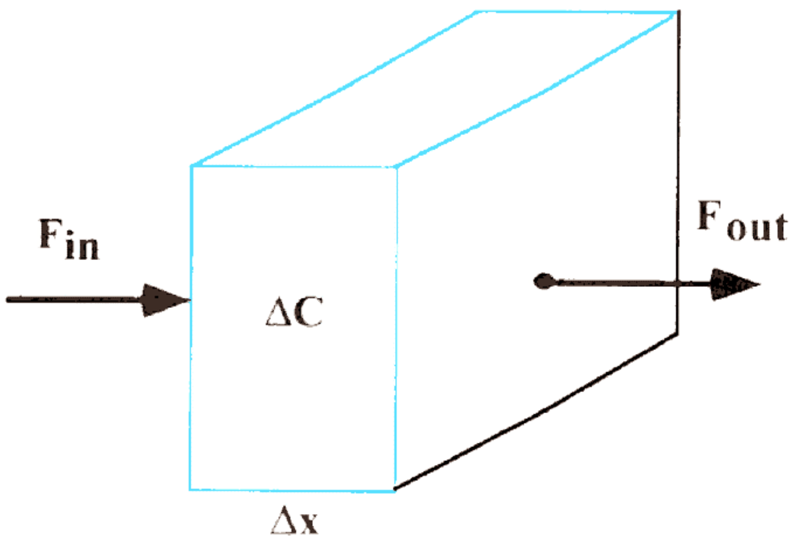
D = diffusion coefficient

Flux– the number of atoms that crosses a plane per unit area per second

When concentration gradient goes to zero, the flow stops.

Fick's Second Law

how the change in concentration in a volume element is determined by the change in fluxes in and out of the volume.



$$\frac{\Delta C}{\Delta t}\Delta x = -(F_{out} - F_{in})$$

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}\left(D\frac{\partial C}{\partial x}\right)$$

$$\frac{dC}{dt} = \frac{Dd^2C}{dx^2}$$

4.3.2 Dopant diffusion coefficients

- Related to the atomic hop rate over an energy barrier and is exponentially activated.

$$D = D^0 \exp\left(\frac{-E_A}{kT}\right)$$

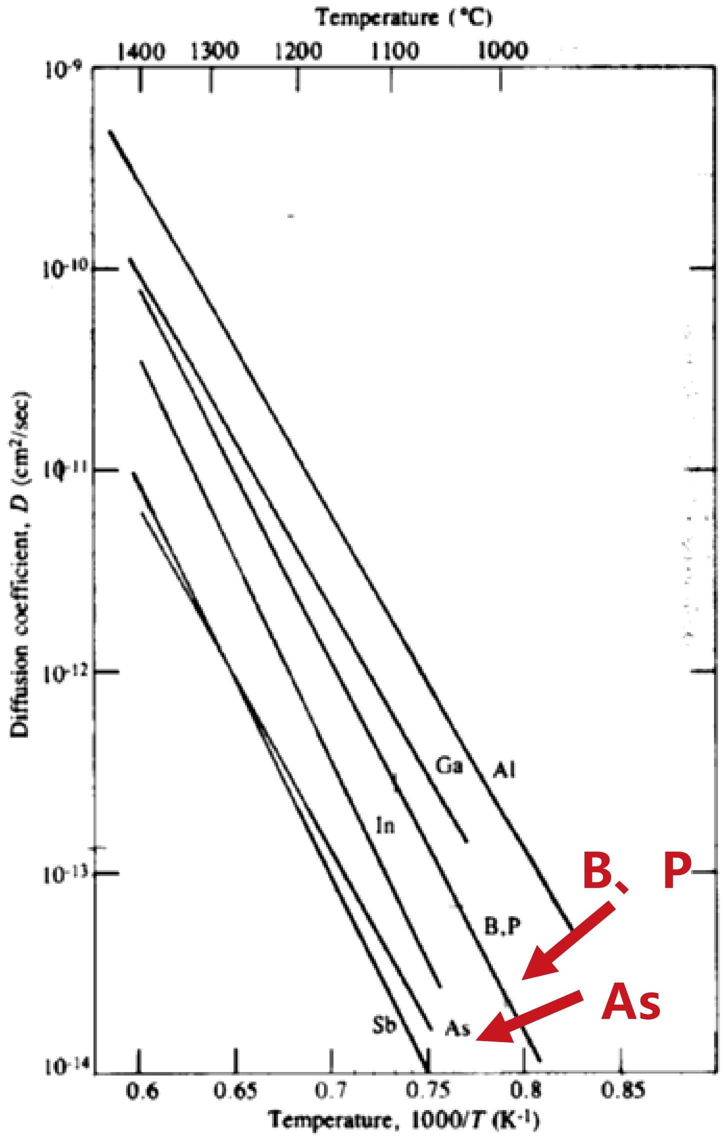
Arrhenius Relationship

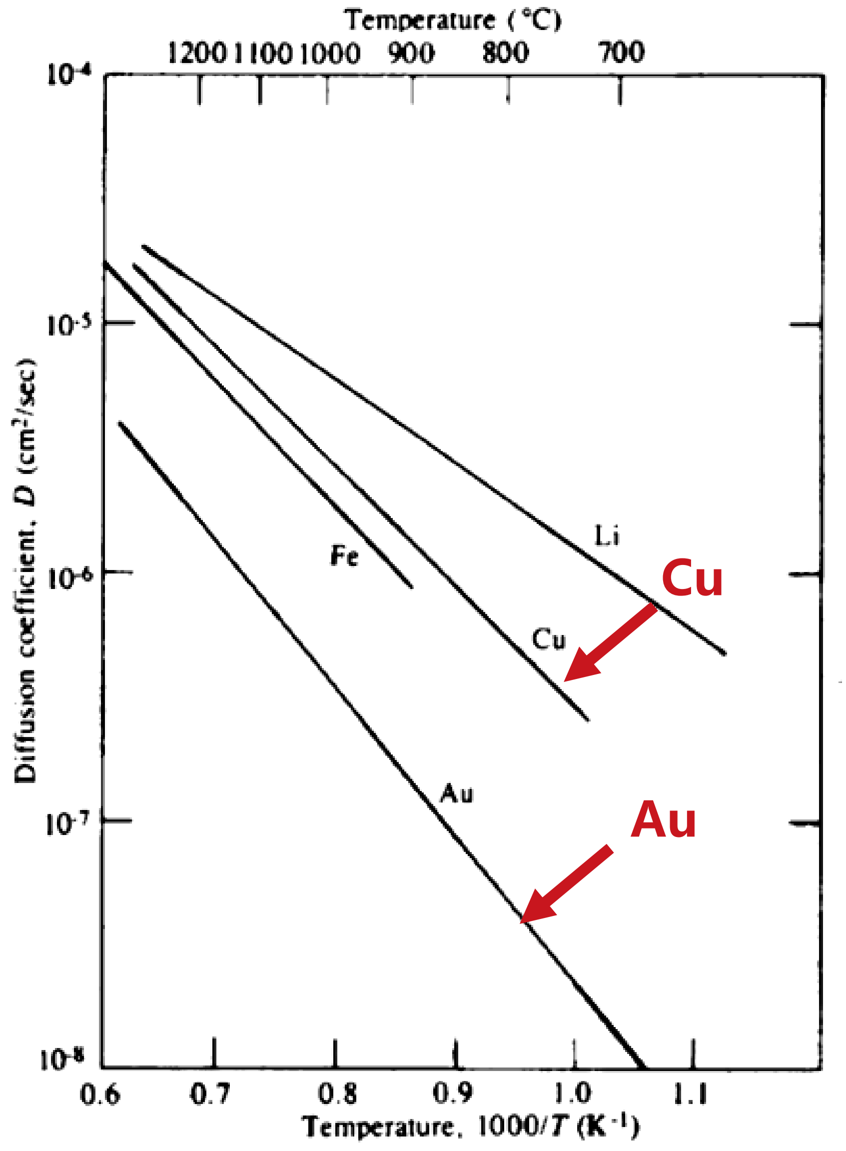
E_A: activation energy
k: Boltzmann's constant
T: absolute temperature

TABLE 4.1 Typical Diffusion Coefficient Values for a Number of Impurities.

Element	D_0(cm²/sec)	E_A(eV)
B	10.5	3.69
Al	8.00	3.47
Ga	3.60	3.51
In	16.5	3.90
P	10.5	3.69
As	0.32	3.56
Sb	5.60	3.95

4.3.2 Dopant diffusion coefficients



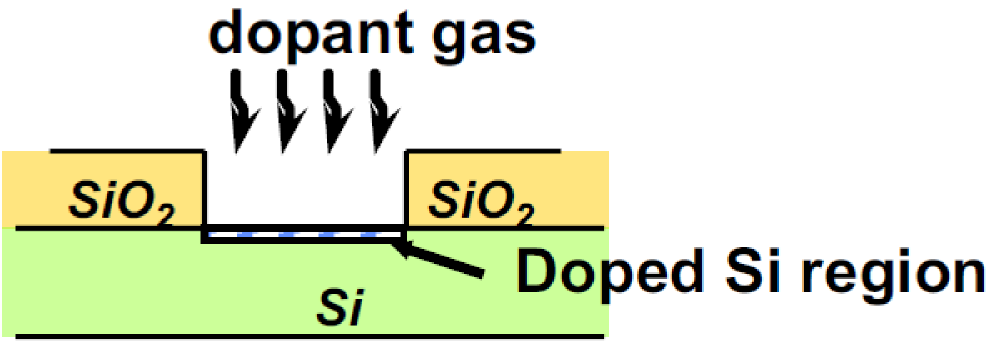


4.3.3 Two step Dopant Diffusion

IC fabrication processes involve a number of different steps

(1) Predeposition

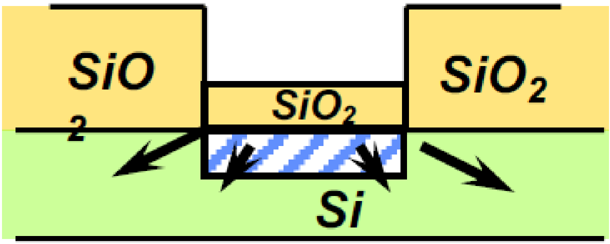
dose control



(2) Drive-in

profile control
(junction depth;
concentration)

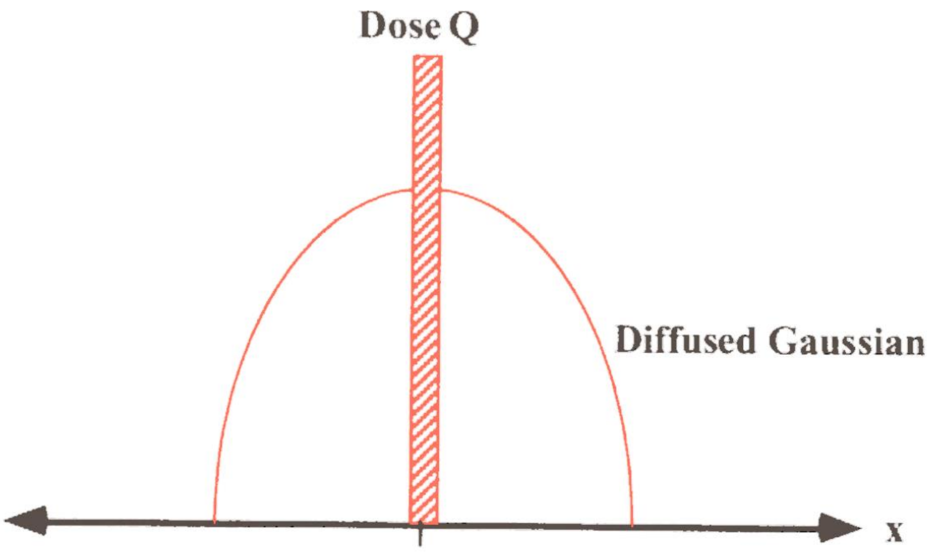
Turn off dopant gas
or seal surface with oxide



Note: Predeposition by diffusion can be replaced by a shallow implantation step.

4.3.4(1) Constant does

Consider a fixed does Q, introduced as a delta function at the origin



Boundary condition:

C→0 at t →0 for x>0

C→∞ at t→0 for x=0

$$\int C(x,t)dx = Q$$

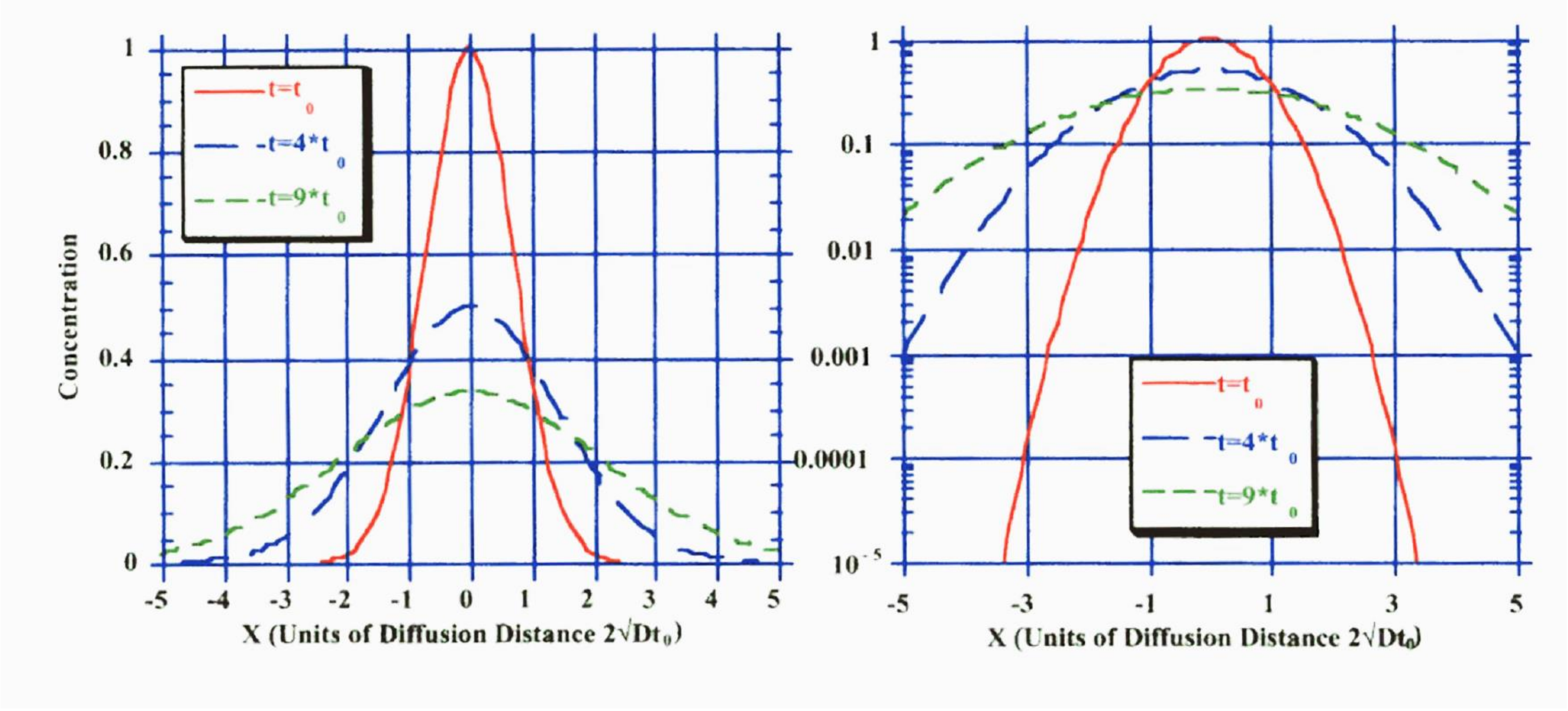
Delta function

$$C(x,t) = C_0 \exp\left[-\left(\frac{x}{2\sqrt{Dt}}\right)^2\right] = \frac{Q}{2\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

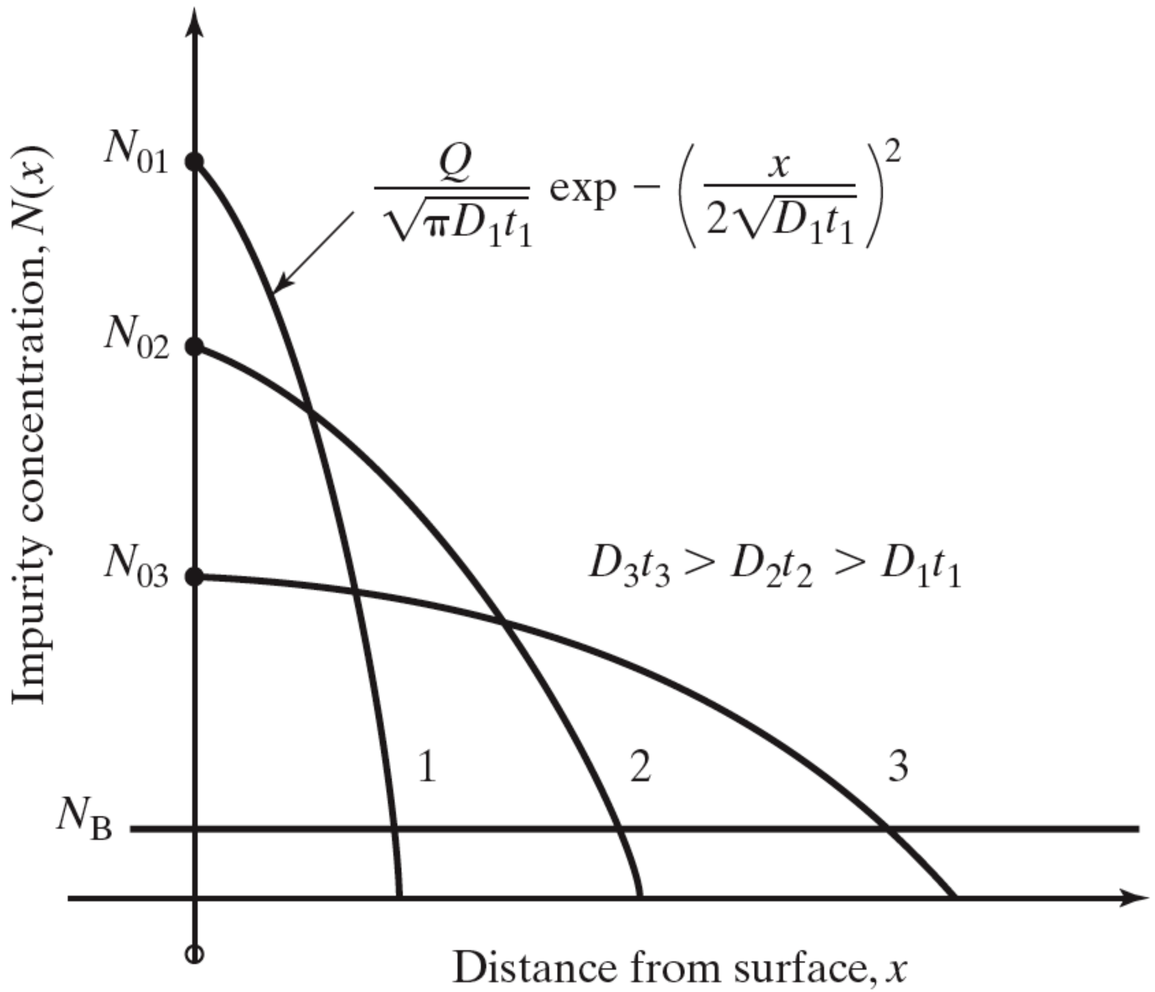
C_0: Surface Concentration

4.3.4(1) Gaussion Diffusion

Time evolution of a Gaussian diffusion profile

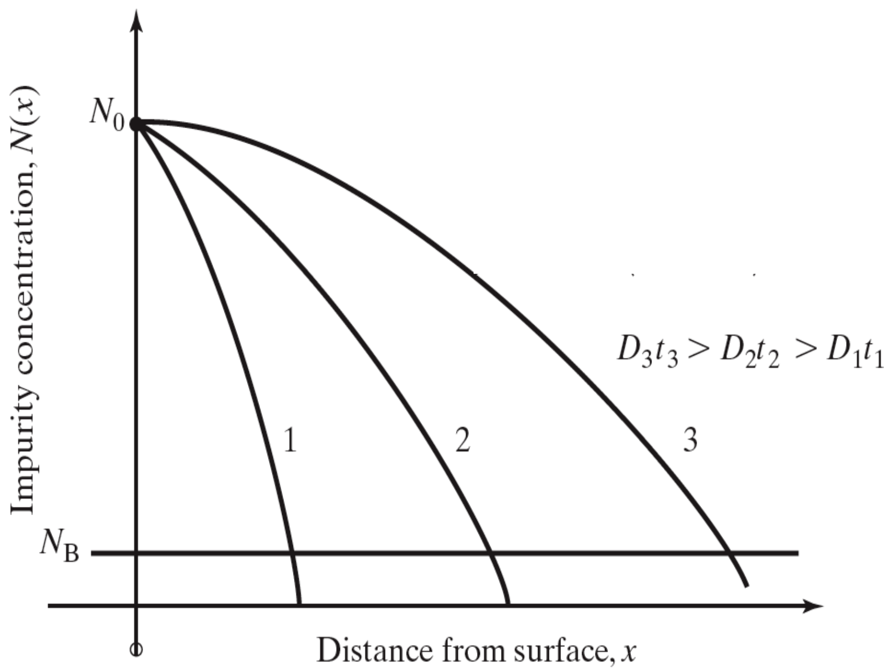


4.3.4(1) Gaussion Diffusion



4.3.4(2) Constant surface concentration

Example: diffusion from a gas ambient into the solid, with the gas concentration above the solid solubility of the dopant



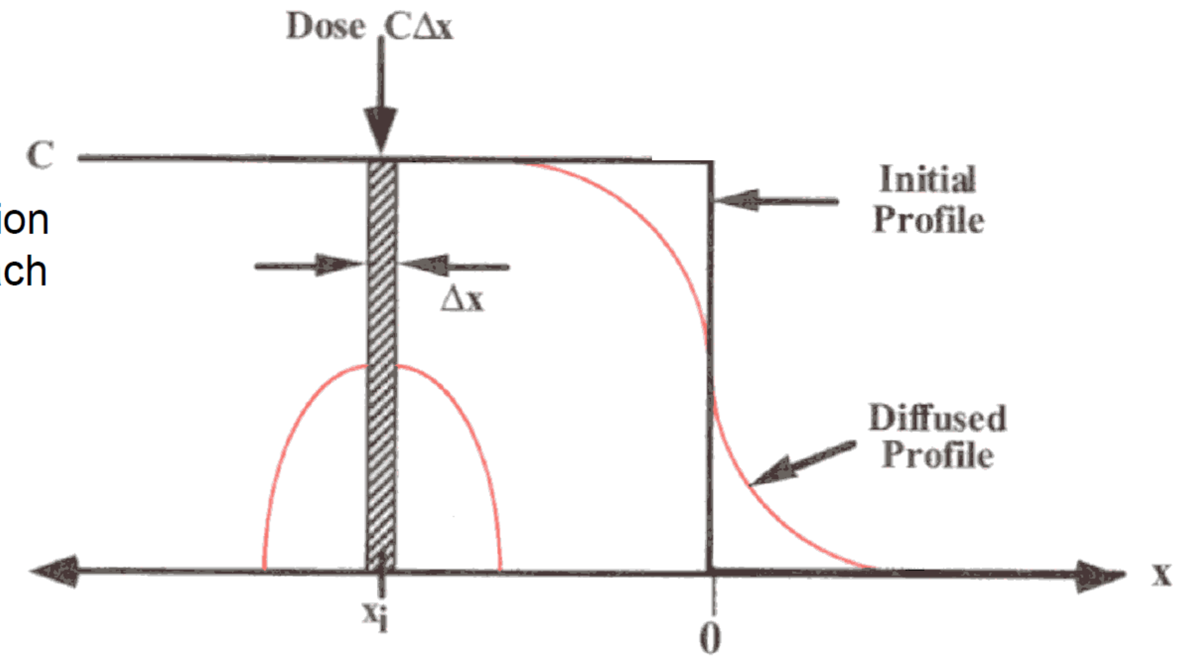
$$C(x,t) = C_0\,\mathrm{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

C_0: Surface concentration
D: Diffusion coefficient
erfc: Complementary Erro function

$$Q = \int_0^{\infty} C(x,t)\,dx = 2\,C_0\sqrt{\frac{Dt}{\pi}}$$

4.3.4(2) Constant surface concentration

Linear superposition of solutions for each of the thin slices:



$$C(x,t)=\frac{Q}{2\sqrt{\pi Dt}}\exp\left(-\frac{x^2}{4Dt}\right)$$

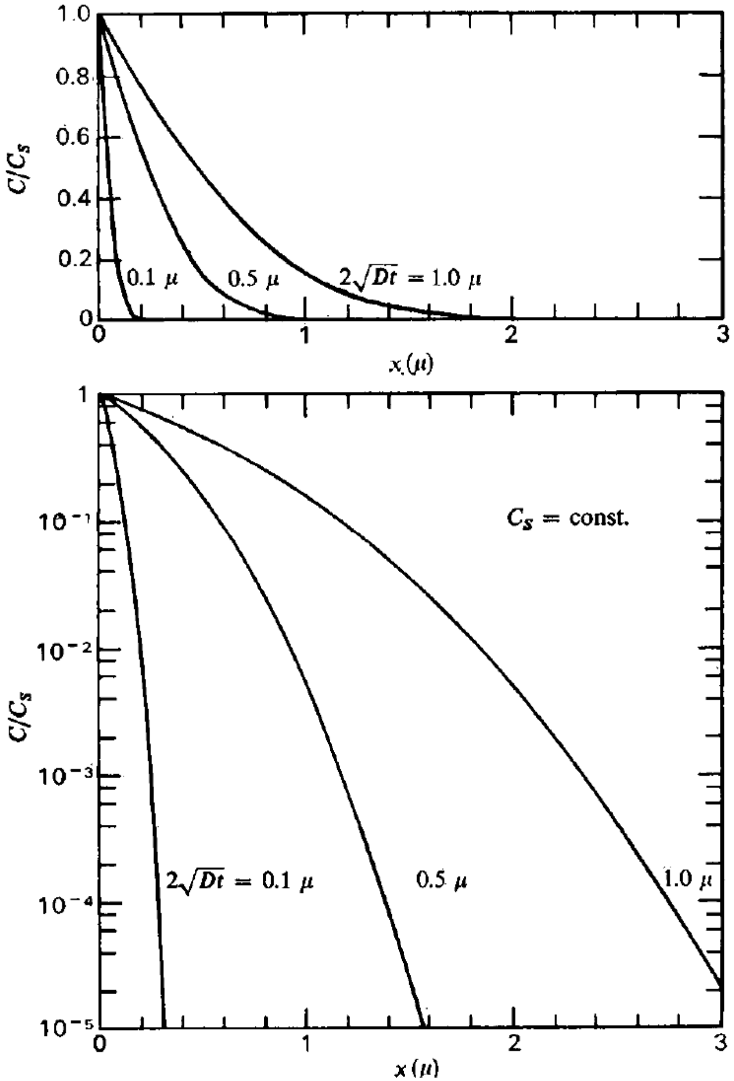
$$C(x,t)=\frac{C}{2\sqrt{\pi Dt}}\sum_{i=1}^{n}\Delta x_i \exp-\frac{(x-x_i)^2}{4Dt}$$

$$\operatorname{erf}(z)=\frac{2}{\sqrt{\pi}}\int_0^z \exp(-\eta^2)d\eta$$

Erro function solution is made up of sum of Gaussian delta function solution

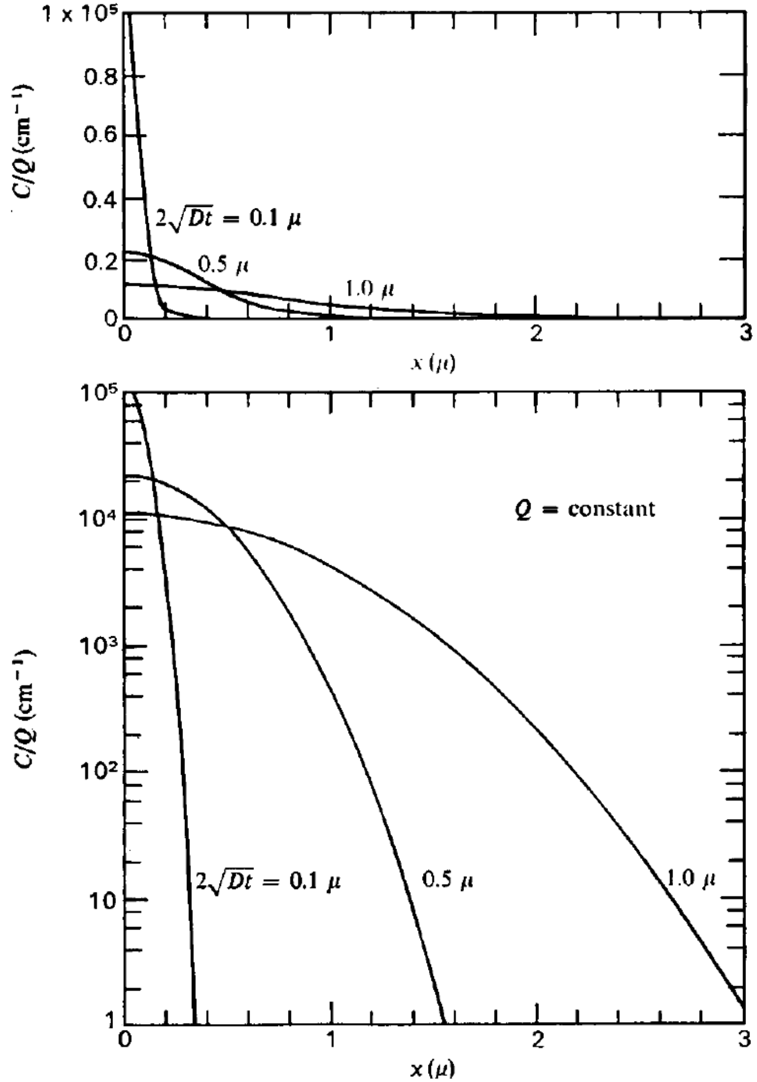
$$C(x,t)=C_0\operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Constant surface conc.



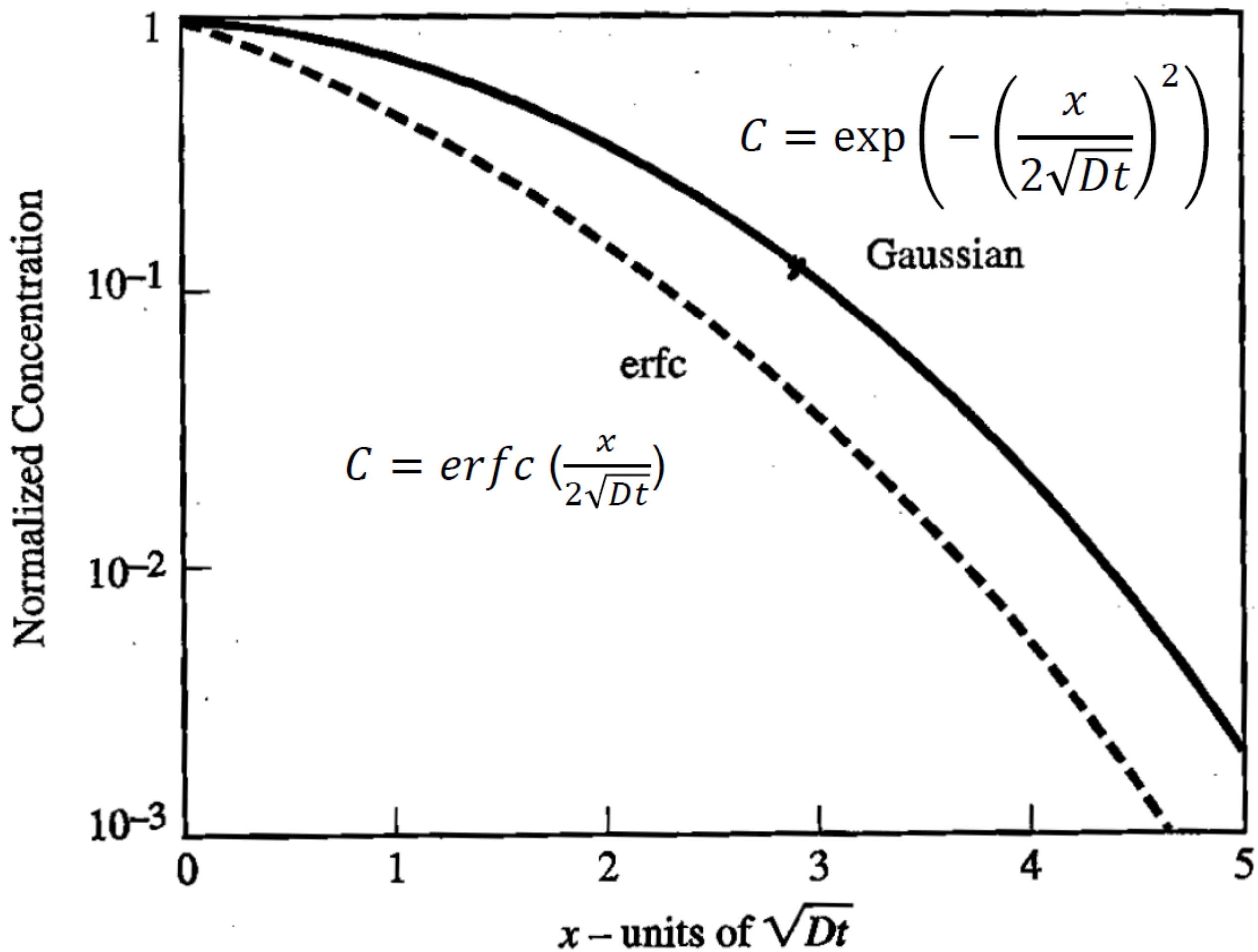
erfc

Constant dose Q



Gaussian

4.3.4 (3) Two kinds of diffusion comparison

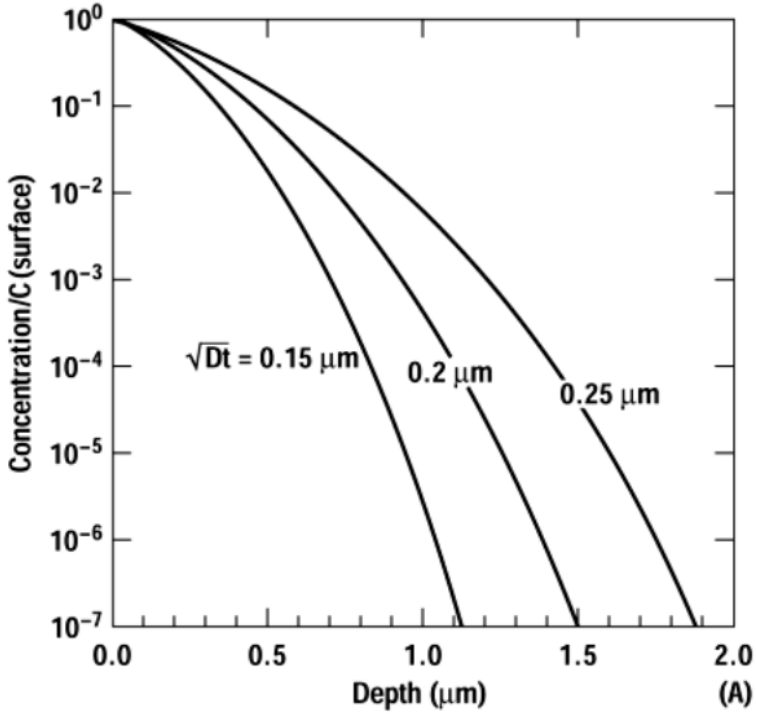


- Similar shape when normalized to the same surface concentration

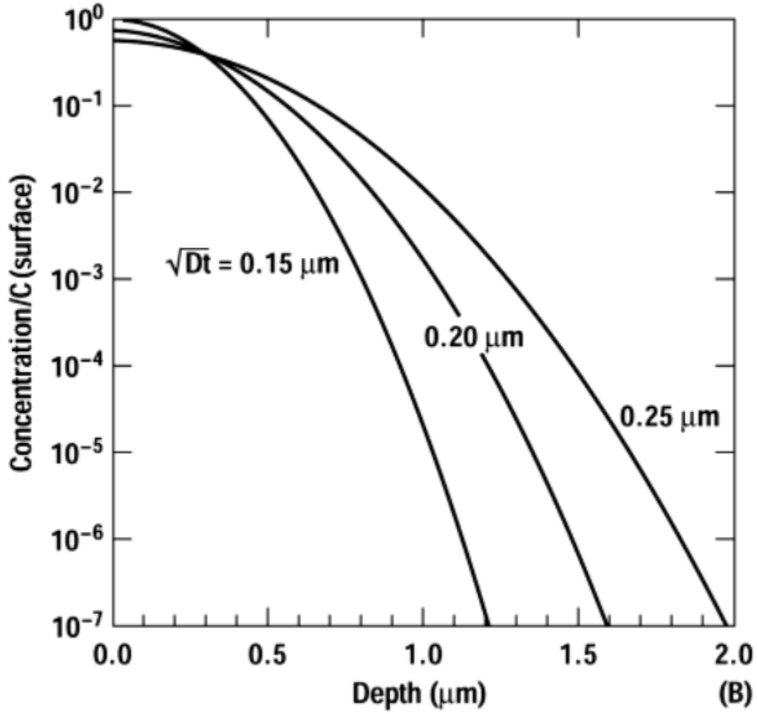
4.3.4 Two step Dopant Diffusion

Normalized Concentration versus Depth

Predeposition

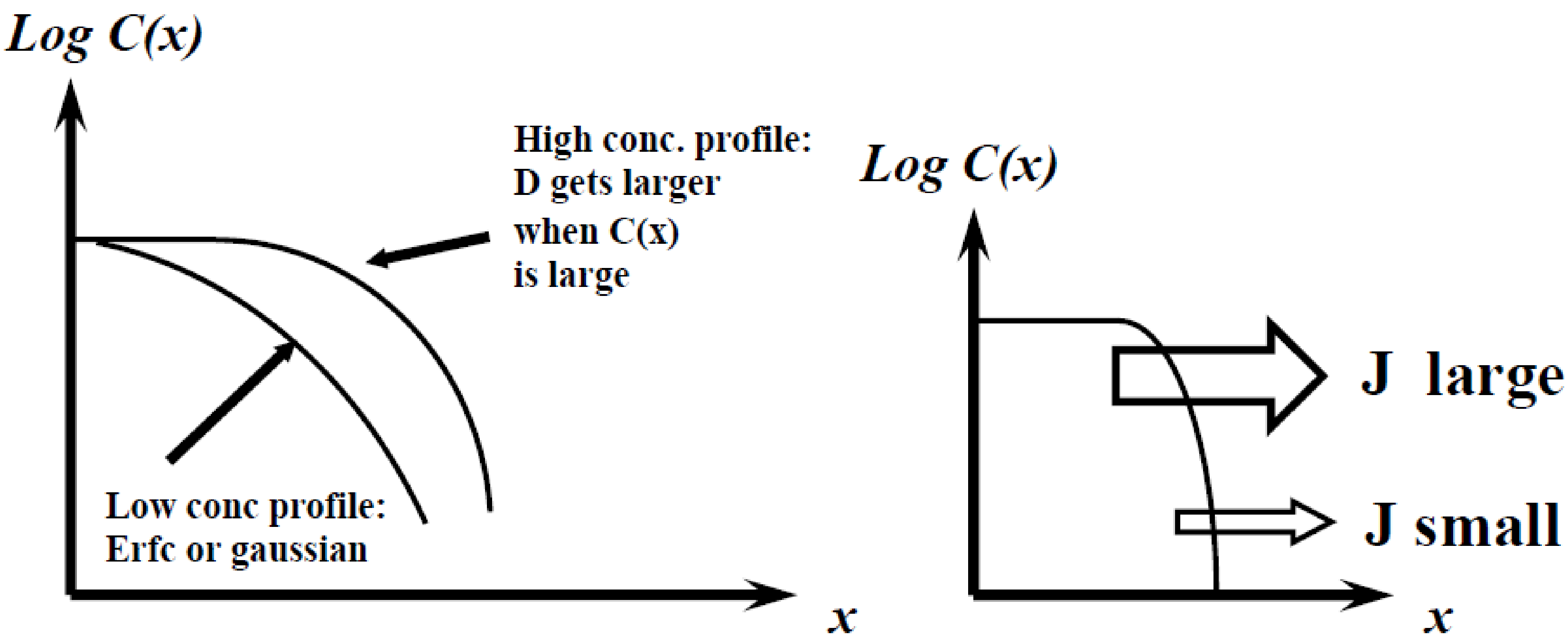


Drive-in



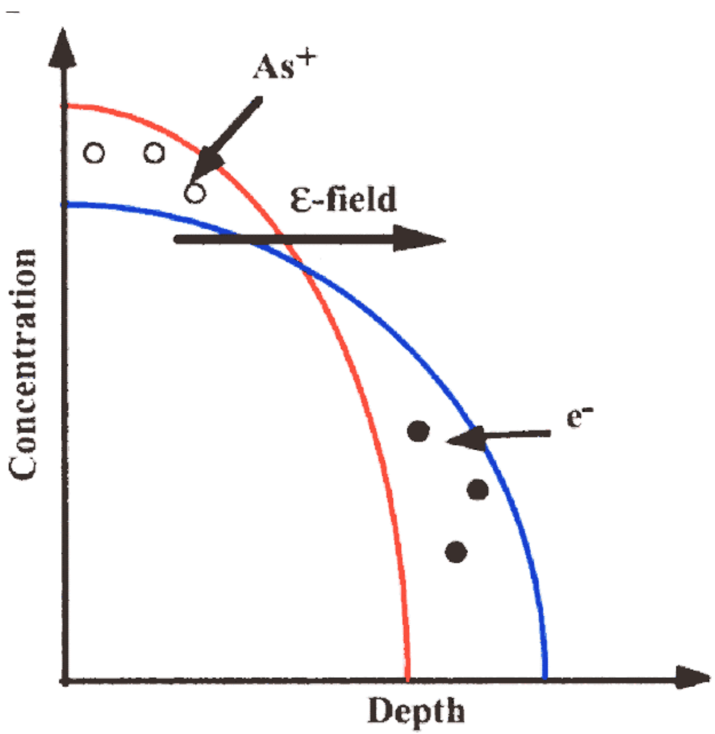
Outline

4.4.1High concentration Diffusion effects



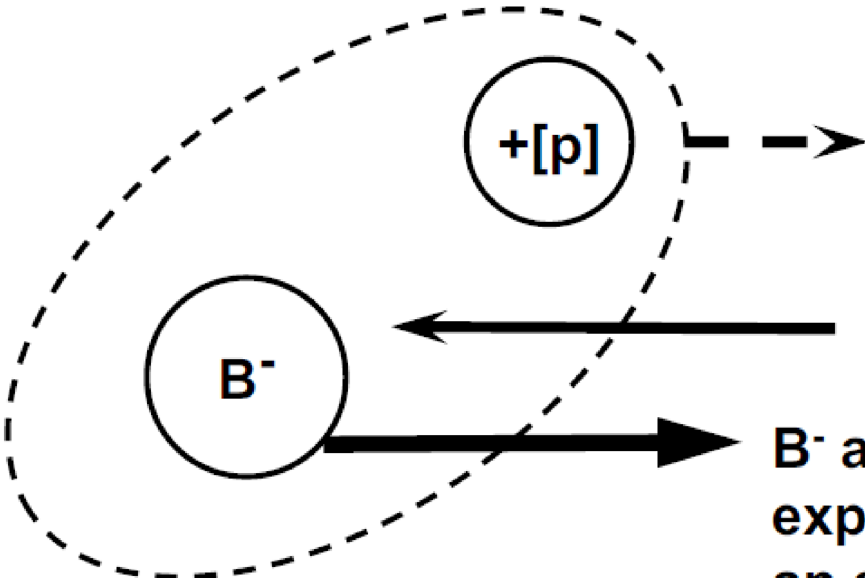
4.4.2 Field enhanced diffusion

When the doping is higher than n_i, Electric filed effects become important



Electrons or holes diffuse ahead of the dopants until they reach a steady-state condition where the drift flux from the internal electric field balances the electron diffusion flux.

4.4.2 Field enhanced diffusion



holes tend to move away due to hole concentration gradient

$E_{build\text{-}in}$

B^- acceptors experience an additional drift force

Enhanced Diffusion for B^- acceptor atoms

4.4.3 Modification of Fick's law

Total impurity atom flux in the presence of both a concentration gradient and an electric field E is

$$F_{total} = F + F' = -\frac{DdC}{dx} + z\mu CE$$

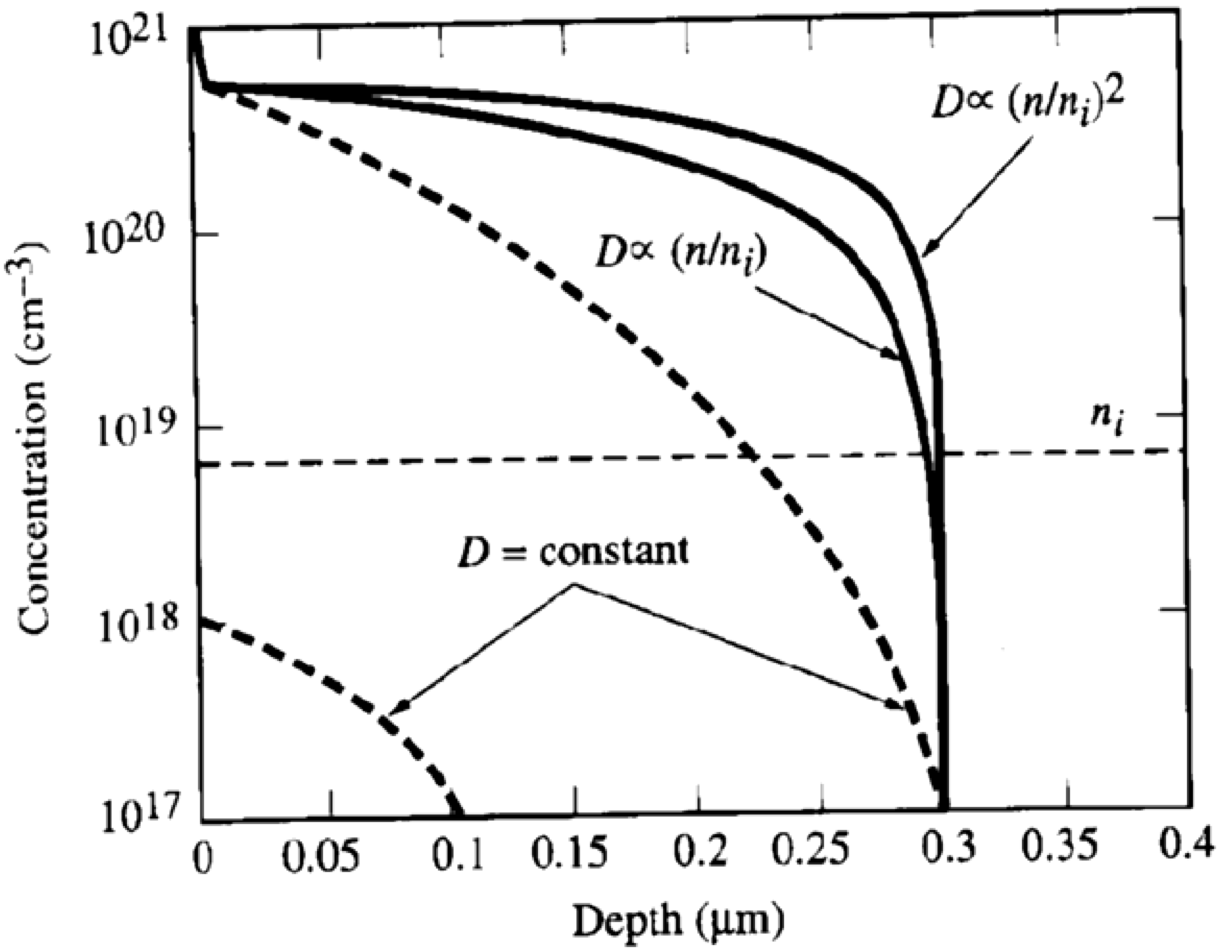
Einstein relation $\mu = D\frac{q}{kT}$

Electric field $E = \frac{\partial\varphi}{\partial x}$ $\varphi = -\frac{kT}{q} ln\frac{n}{n_i}$

$$F_{total} = -\frac{DdC}{dx} - \mathrm{DC}\frac{\partial}{\partial x}\ln\frac{n}{n_i}$$

4.4.4 Dopants profiles

- **Example : High Concentration Arsenic diffusion profile becomes "box--like"**



4.4.5 Summary of high concentration effect

- If doping conc $< n_i$:
  - Use constant diffusivity solutions
  - (profile is erfc or half-gaussian)

- If doping conc $> n_i$:
  - Solution requires numerical techniques

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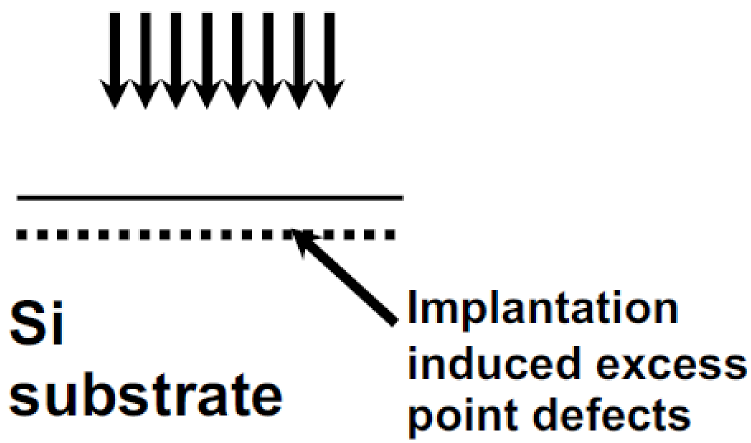
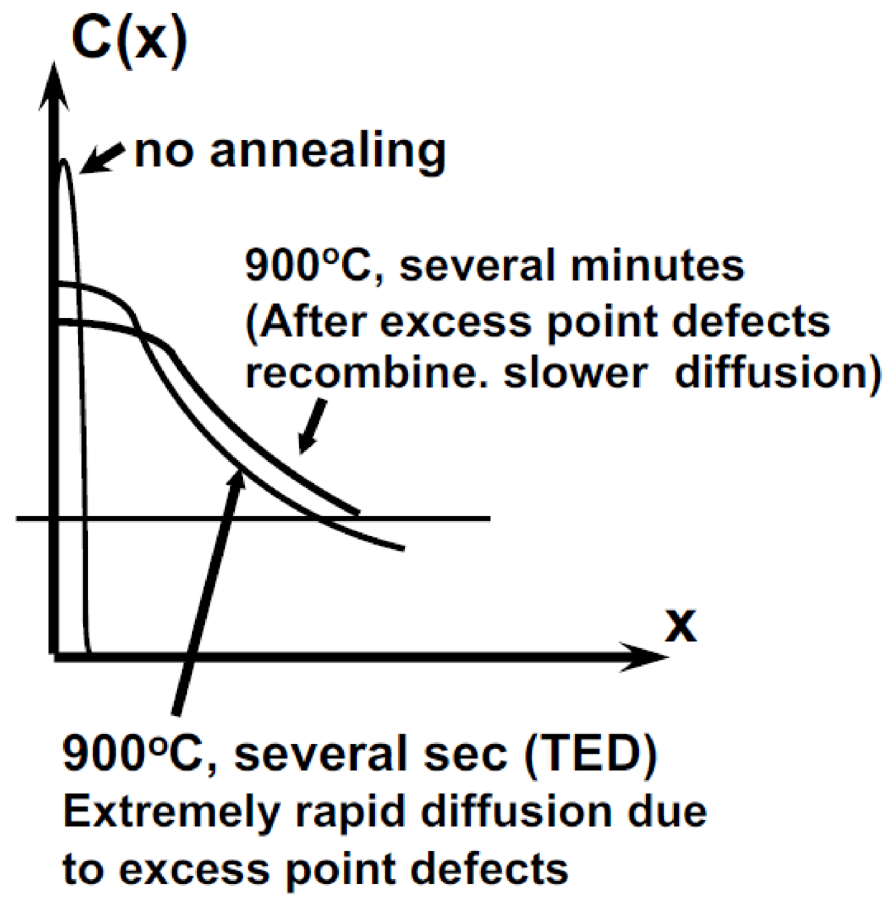
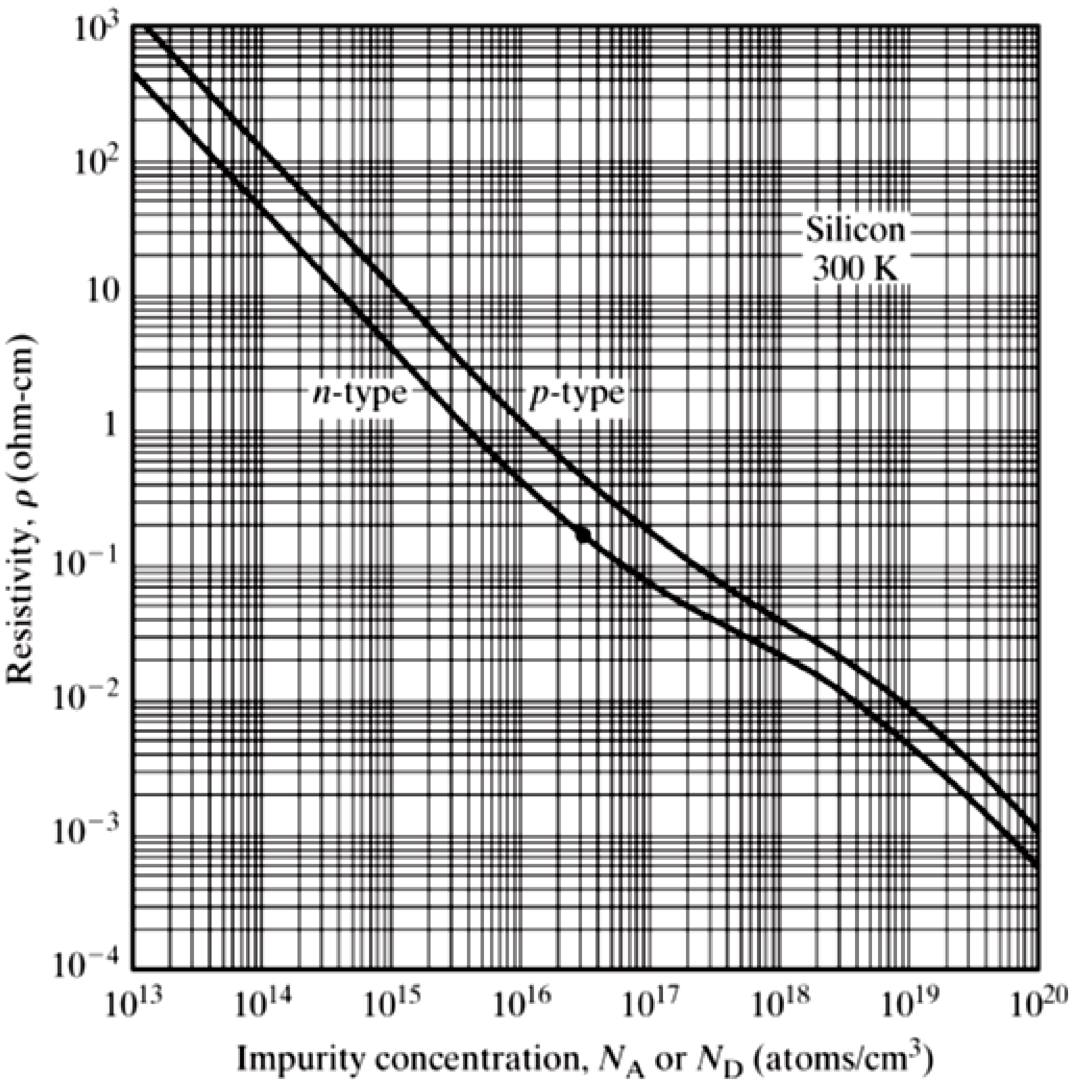
Dopant Implantation





Implantation creates large number of excess Si interstitials and vacancies. (1000X than thermal process). After several seconds of annealing, the excess point defects recombine.

4.5.2 Resistivity Vs. Doping



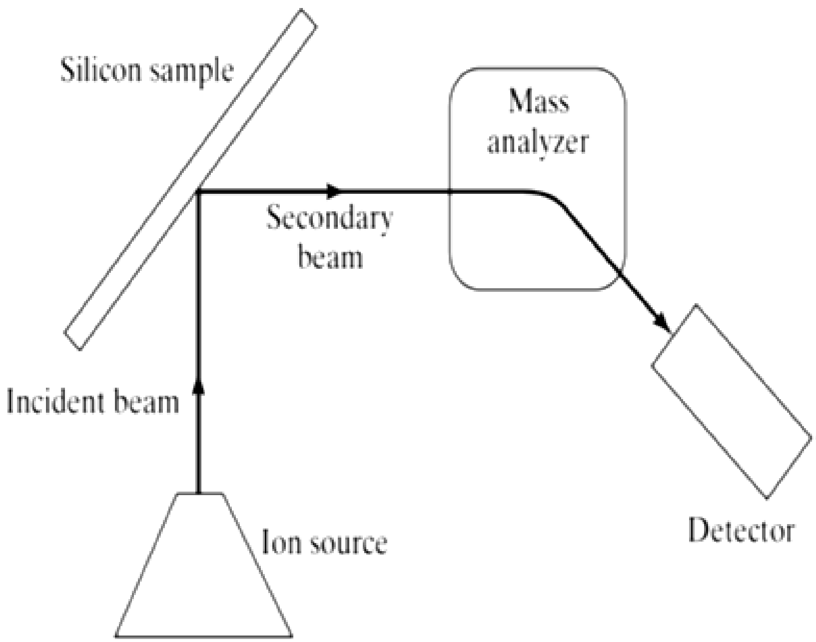
$$\rho = \sigma^{-1} = \left[q\left(\mu_n n + \mu_p p\right)\right]^{-1}$$

$$\text{n-type}: \ \rho \cong \left[q\mu_n\left(N_D - N_A\right)\right]^{-1}$$

$$\text{p-type}: \ \rho \cong \left[q\mu_p\left(N_A - N_D\right)\right]^{-1}$$

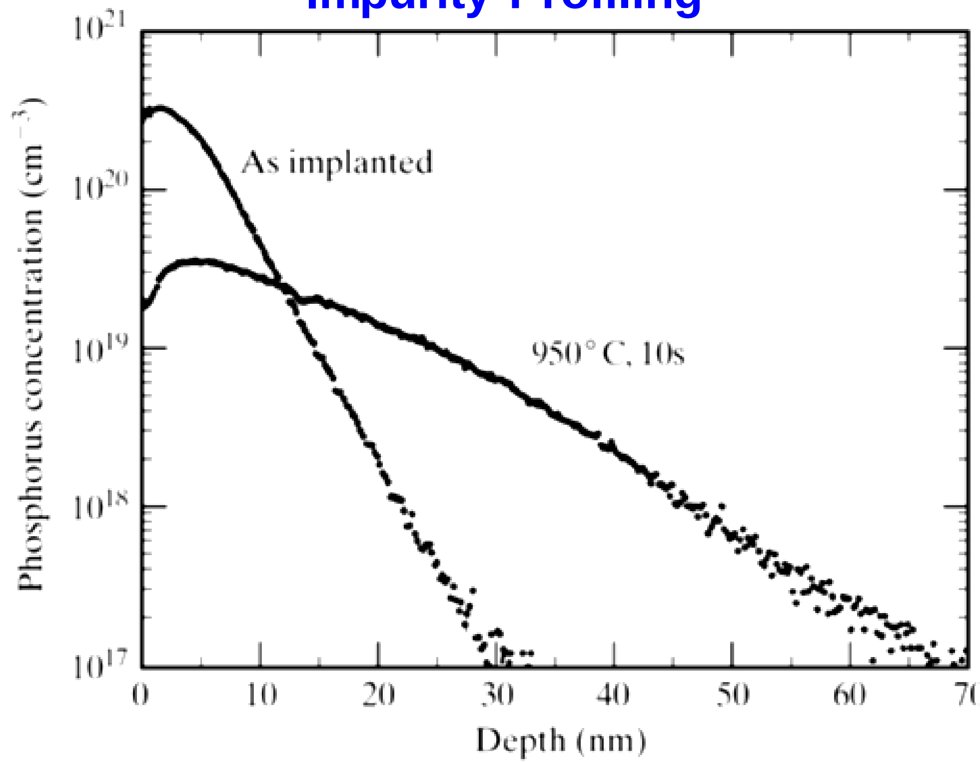
4.5.3 SIMS for impurity profiles

Secondary Ion Mass Spectroscopy (SIMS)



(a)

Impurity Profiling



(b)

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4.6.1 Summary

- **Why is diffusion important?**

Determine impurity concentration and resistivity

- **Basics of diffusion**

Definition, solubility, mechanism

- **Fick's law for diffusion** $F=-\frac{DdC}{dx}$ $\qquad \frac{dC}{dt}=\frac{Dd^2C}{dx^2}$

First and second law

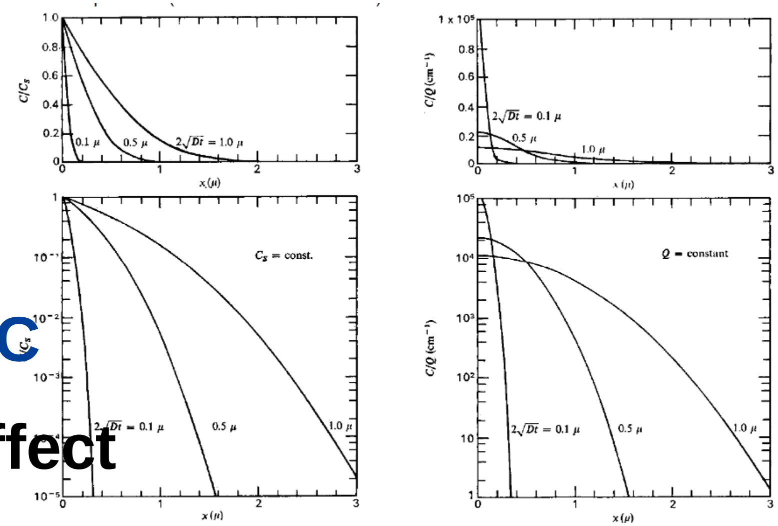
- **Two types of diffusion**

Gaussion and error function

Predeposition, drive-in in actual IC

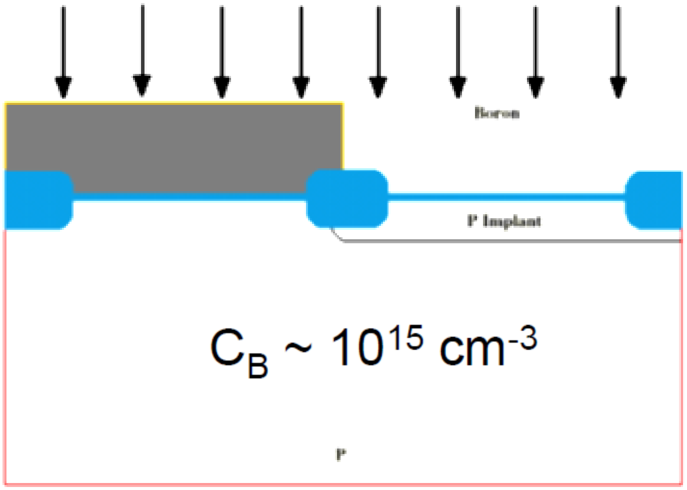
- **High concentration diffusion effect**

Why? How?



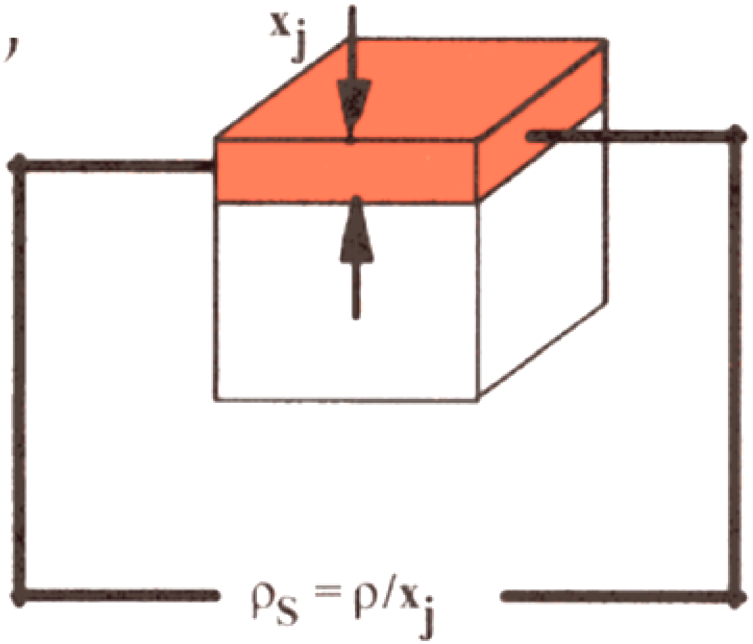
4.6.2 Homework

Design the Drive-in Conditions (i.e. the temperature and time) for a CMOS P-well



Want sheet resistance =900 ohm/square
And Xj = 3 um

4.6.2 Homework



The sheet resistance of a shallow junction with uniform doping

$$R = \frac{\rho}{x_j} \ \Omega/\text{Square} \equiv \rho_S$$

In general: $$\rho(x) = \frac{1}{q\, n(x)\, \mu[n(x)]}$$

(ohm-cm)

For a non-uniformly doped layer

$$\rho_s = \frac{\overline{\rho}}{x_j} = \frac{1}{q \int_0^{x_j} [n(x) - C_B]\, \mu[n(x)]\, dx}$$

This equation has been integrated numerically (by Irvin) for several analytical profiles (e.g. Gaussian and erfc profiles)

$$\rho_S = 900\ \Omega/\text{square}$$

$$x_j = 3\ \mu\text{m}$$

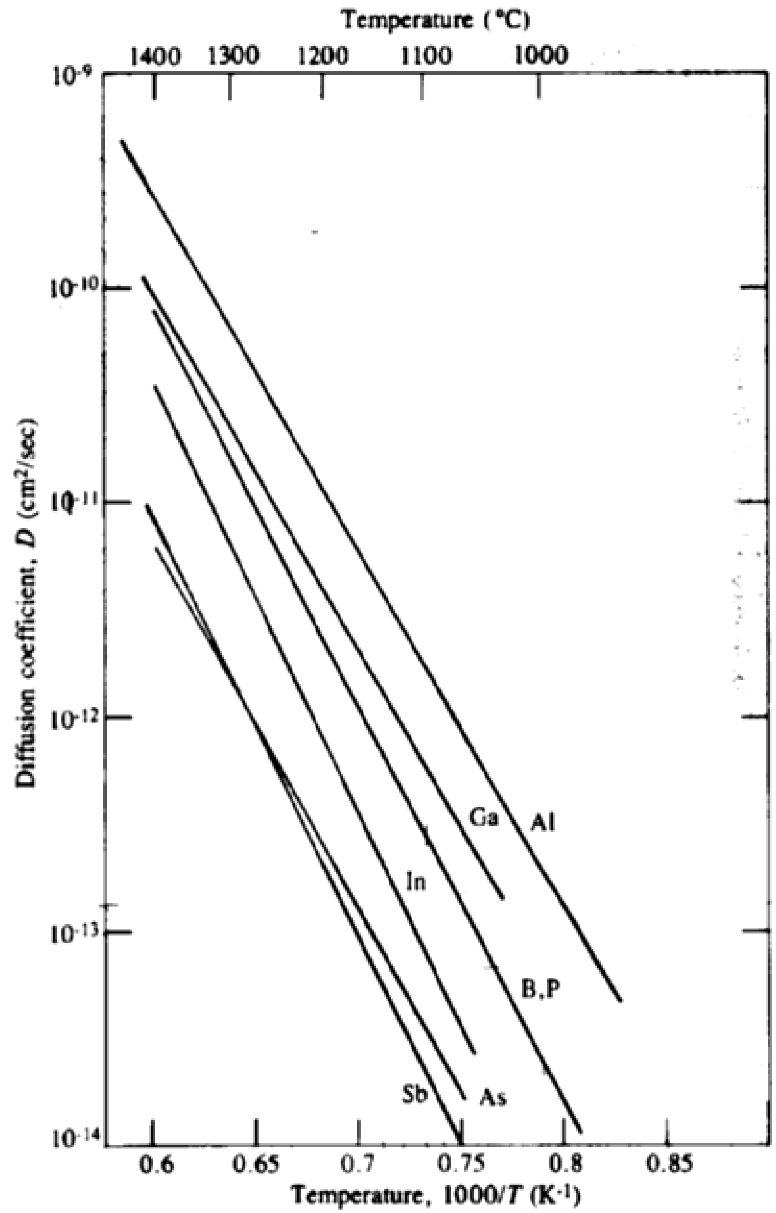
$$C_B = 1\text{x}10^{15}\ \text{cm}^{-3}\ \text{(substrate concentration)}$$

- **The average conductivity of the layer is**

$$\bar{\sigma} = \frac{1}{\rho_S x_j} = \frac{1}{(900\Omega/\text{sq})(3\times10^{-4}\text{cm})} = 3.7\ (\Omega\cdot\text{cm})^{-1}$$

- **From Irvin's curve we obtain** (assuming Gaussian)

$$C_S \approx 4\times10^{17}/\text{cm}^3$$



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